Veritas Engineering Blueprint

1. Project Overview

Veritas is an AI-powered research verification platform that retrieves scientific papers, extracts evidence, verifies claims, and produces cited answers.

2. Architecture

User -> FastAPI -> Retrieval -> PDF Parsing -> Chunking -> Embeddings -> Qdrant -> Hybrid Retrieval -> NLI Verification -> Agents -> Next.js Dashboard.

3. Tech Stack

FastAPI, PostgreSQL, Docker, PyMuPDF, Sentence Transformers, Qdrant, BM25, Cross Encoder, Neo4j, Hugging Face, LangGraph, Next.js.

4. Folder Structure

backend/, frontend/, retrieval/, parser/, embeddings/, verification/, knowledge_graph/, agents/, database/, docker/.

5. Phase 1

Build FastAPI backend, Docker Compose, PostgreSQL, health endpoint, /research endpoint, DB models, migrations, testing.

6. Phase 2

Retrieve papers from arXiv and PubMed, download PDFs, parse text, clean and chunk documents.

7. Phase 3

Generate embeddings, store vectors in Qdrant, implement semantic search and hybrid retrieval with BM25 + reranker.

8. Phase 4

Extract entities and claims, build Neo4j knowledge graph, visualize relationships.

9. Phase 5

Use NLI models to classify support, contradiction or neutral evidence. Add planner, retriever, verifier and synthesizer agents.

10. Frontend

Build Next.js dashboard with search, progress updates, evidence viewer, graph viewer and report generation.

11. APIs

POST /research, GET /papers, POST /verify, GET /graph, GET /status.

12. Milestones

Backend ✓ Retrieval ✓ Semantic Search ✓ Knowledge Graph ✓ Verification ✓ Agents ✓ Frontend ✓ Deployment.